

21-602: Introduction to Set Theory I

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Chapter 1

A Recap of Undergraduate Set Theory

Set Theory can be viewed as a field of mathematics that encompasses all other mathematics. We can view set theory as beginning at the empty set $\emptyset$, also denoted V_0 . We can construct subsequent sets $V_1, V_2, V_3, \dots$ by taking power sets, ie, $V_1 = \mathcal{P}(V_0)$, $V_2 = \mathcal{P}(V_1)$, and so on. We can then define V_ω to be the union of all V_n , where $n < \omega$. We can then define $V_{\omega+1} = \mathcal{P}(V_\omega)$, and so on. All of analysis, the field of mathematics Davoud Cheraghi once described as the “rigorous study of infinite constructions”, is essentially about $V_{\omega+1}$. In this way, set theory encompasses analysis, for example.

We call the hierarchy

$$V_0 \subset V_1 \subset \dots \subset V_\omega \subset V_{\omega+1} \subset \dots$$

the **universe of sets**. Indeed, this is often represented diagrammatically as a V that contains each of these V s, with V_0 at the bottom, V_1 above V_0 , and so on.

There is a problem with the universe of sets: *the power set operation is too strong*. That is, when we take the power set of a set, it “throws too many elements in”. This leads to questions like the *continuum hypothesis*, which asks the question of whether $2^{\aleph_0} = |\mathbb{R}| = |V_{\omega+1}|$ is equal to $\aleph_1$.

Something we will study quite closely in this course is a slightly different version of this universe, known as **Gödel’s Constructible Universe**. This is a universe of sets that is not built up from $\emptyset$ using the power set, but using a *different* operation $\mathcal{D}$, defined as follows: for a set A , viewing

it as a first-order structure $(A; \in)$,

$$\mathcal{D}(A) := \{x \mid x \text{ is a 1st-order definable structure over } (A; \in) \text{ using parameters from } A\}$$

We can show that for finite sets, $\mathcal{D}$ agrees with $\mathcal{P}$. Moreover, we can show that for a countable set A , $\mathcal{D}(A)$ is countable as well. The point is that if we then define a universe

$$(L_0 = \emptyset) \subset (L_1 = \mathcal{D}(L_0)) \subset \cdots \subset \left(L_\omega = \bigcup_{n < \omega} L_n \right) \subset \cdots$$

we not only have agreement between L_n and V_n for all $n \leq \omega$, but we can build a model of ZFC in which the continuum hypothesis is true!

This is something we will explore in great detail in this course.

One convention first, which we use throughout the chapter.

Convention. When we write $\sup$, we mean “union over”. Just like in Lean.

We will begin by reviewing the ZFC axioms.

1.1 The Zermelo-Fraenkel Axioms and the Axiom of Choice

We begin by reviewing the Zermelo-Fraenkel Axioms of Set Theory. We work in a first-order language with only one relation symbol, denoted $\in$. In principle, we would want to distinguish between the symbol in the language and its interpretation in any model, but we will not do this in practice. In situations where we really want to be pedantic, we will denote the symbol $\dot{\in}$ and the interpretation $\in$.

We begin with the Axiom of Extensionality.

1.1.1 Extensionality

ZFC Axiom 1 (The Axiom of Extensionality).

$$\forall A \forall B [(\forall x (x \in A \leftrightarrow x \in B)) \rightarrow (A = B)]$$

We define the $\subseteq$ symbol, used infix as $A \subseteq B$, to be shorthand for the formula $\forall x (x \in A \rightarrow x \in B)$. This is not a symbol in the language of set theory; it is merely a convenient abbreviation that we will use hereafter without further discussion.

ZFC Axiom 1 essentially tells us that

$$\forall A \forall B ((A \subseteq B \wedge B \subseteq A) \rightarrow A = B)$$

(This is provably equivalent to ZFC Axiom 1.)

We now define the Axiom Scheme of Comprehension.

1.1.2 Comprehension

First, some notation.

Notation. We will denote finite tuples using overlines, so that $\bar{y}$ refers to $(y_0, \dots, y_{n-1})$. We will typically use bars to bundle together variables.

ZFC is an infinite theory. We call the following an Axiom “Scheme” to underscore the fact that it is really an infinite family of axioms bundled together.

ZFC Axiom 2 (The Axiom (Scheme) of Comprehension). Let $x, \bar{y}$ be free variables. For all formulae $\varphi(x, \bar{y})$, the following is an axiom:

$$\forall A \forall \bar{y} \exists B \forall x [x \in B \leftrightarrow (x \in A \wedge \varphi(x, \bar{y}))]$$

Intuitively, this means we can define

$$B = \{x \in A \mid \varphi(x, \bar{y})\}$$

This is an axiom scheme because we can increase the arity of φ : we don't know how many variables y_i are bundled together in $\bar{y}$, only that for any given statement of the axiom, there are finitely many y_i .

Note that ZFC Axiom 1 tells us that we can replace the existential quantifier for B in ZFC Axiom 2 with an existence and uniqueness quantifier without changing its meaning.

Before proceeding further, we define the empty set. In first order logic, structures are required to have non-empty universes. Thus, for every structure (A, E) in the language of set theory, with $E \subseteq (A \times A)$ representing equality, we must have

$$(A, E) \models \exists x (x = x)$$

The Completeness Theorem then tells us that

$$\vdash \exists x (x = x)$$

ie, that the formula $\exists x (x = x)$ *must be a theorem* in the language of sets. Hence, ZFC Axiom 1 and ZFC Axiom 2 tell us that

$$\vdash \exists! A \forall x (x \notin A)$$

In other words, it is a theorem in the language of sets that there is a unique set that contains no members whatsoever. This unique set is denoted $\emptyset$, and is called the **empty set**. Its existence (and uniqueness) is not a distinct axiom, but a direct consequences of ZFC Axiom 1 and ZFC Axiom 2.

Next, we give the axiom of pairing.

1.1.3 Pairing

ZFC Axiom 3 (The Axiom of Pairing).

$$\forall x \forall y \exists A (x \in A \wedge y \in A)$$

ZFC Axiom 3 essentially gives us a way of constructing, for any x and y , the unique set $A = \{x, y\}$

with the property that

$$\forall z (z \in A \leftrightarrow (z = x) \vee (z = y))$$

In particular, it allows us to construct, for any x , the set $\{x, x\}$, which, by ZFC Axiom 1, is exactly $\{x\}$. In other words, it tells us that we can stick any x into a set that only contains x .

This allows us to construct pairwise unions, but not arbitrary unions.

1.1.4 Unions

Now, we state the axiom of unions.

ZFC Axiom 4 (The Axiom of Unions). If we denote by $\mathcal{F}$ a family of sets,

$$\forall \mathcal{F} \exists B \forall A [A \in \mathcal{F} \rightarrow A \subseteq B]$$

As usual, we can use ZFC Axiom 1, ZFC Axiom 2 to show that

$$\bigcup \mathcal{F} = \{x \mid \exists A \subseteq \mathcal{F} (x \in A)\}$$

is well (and uniquely) defined.

Next comes the power set axiom.

1.1.5 Power Sets

ZFC Axiom 5 (The Axiom of Power Sets).

$$\forall A \exists \mathcal{F} \forall X [X \subseteq A \rightarrow X \in \mathcal{F}]$$

Again, ?? 1?? 2 tell us that this is equivalent to

$$\forall A \exists ! \mathcal{F} \forall X (X \in \mathcal{F} \leftrightarrow X \subseteq A)$$

This justifies the definition

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

Next comes the second (and last) axiom scheme in ZFC, the axiom scheme of replacement.

1.1.6 Replacement

ZFC Axiom 6 (The Axiom (Scheme) of Replacement). For each formula $\varphi(x, y, \bar{z})$ that does not contain B as a free variable, the following is an axiom:

$$\forall A \forall \bar{z} [[\forall x \in A \exists! y \varphi(x, y, \bar{z})] \rightarrow [\exists B \forall x \in A \exists y \in B \varphi(x, y, \bar{z})]]$$

We can unpack this slightly complicated formula using ZFC Axiom 1 and ZFC Axiom 2: it essentially tells us that we can define

$$\{y \mid \exists x \in A (\varphi(x, y, \bar{z}))\}$$

ie, if for every x , there is a unique witness y for which the formula $\varphi(x, y, \bar{z})$ holds, we can collect all these witnesses into a set that is contained B . In particular, “the number of such witnesses is *small enough* that it *can be collected into a set*.”

Consider the case where $\varphi(x, y, \bar{z})$ defines a function with domain A . If we can define $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$ (which we have yet to show we can do, but let’s say we can), then we can put

$$f = \{(x, y) \in A \times B \mid \varphi(x, y, \bar{z})\}$$

which is a set by ZFC Axiom 2. Effectively, ZFC Axiom 6 tells us that *if the domain of a function is a set, then so is its range*. ZFC Axiom 6 is an essential step in the process of actually *defining* functions (or showing that this is something we can *do*).

One situation in which ZFC Axiom 6 is used (and will be used once sufficiently many axioms are stated) is in the situation we talked about at the start of Chapter 1. We’ll see that there is a formula $\varphi(n, S)$ so that $\varphi(n, S) \leftrightarrow (n \in \omega \wedge S = V_n)$. Then, ZFC Axiom 6 would imply that

$$\langle V_n \mid n < \omega \rangle$$

really is a sequence. This allows us to define $V_\omega = \bigcup_{n < \omega} V_n$ and continue.

1.1.7 Foundation

ZFC Axiom 7 (The Axiom of Foundation).

$$\forall A (A \neq \emptyset \rightarrow \exists x \in A \forall y \in A y \notin x)$$

We now state the most interesting of the ZFC axioms: the Axiom of Choice.

1.1.8 Choice

ZFC Axiom 8 (The Axiom of Choice - (AC)).

$$\forall \mathcal{F} \exists c [\forall A \in \mathcal{F} (A \neq \emptyset \rightarrow A \in \text{dom}(c) \wedge c(A) \in A)]$$

where $\mathcal{F}$ is a family of sets, c is a function, and $\text{dom}(c)$ is the domain of c . We call c a **Choice Function on $\mathcal{F}$** .

The reason we state the Axiom of Choice at this point in our discussion is that modulo the ZFC Axioms that we have seen so far, the Axiom of Choice is equivalent to each of the following.

1. $\forall A \exists R \subseteq A \times A$ (A, R) is a well-ordering.
2. $\forall A \exists \alpha \exists f : A \xrightarrow{\sim} \alpha$, where α is an ordinal.

Note that none of the above axioms imply the existence of the natural numbers, without which virtually none of the mathematics we know and love would be possible. We therefore come to the axiom of infinity, which posits the existence of infinite sets, the ‘smallest’ of which is the natural numbers.

1.1.9 Infinity

ZFC Axiom 9 (The Axiom of Infinity).

$$\exists I (\emptyset \in I \wedge \forall x \in I (x \cup \{x\} \in I))$$

A set I as above is called an **inductive set**, and it is an axiom of ZFC that such a set exists.

It is possible to prove that given the existence of inductive sets (as posited above), there is a

unique inductive set ω such that for all inductive sets J , $\omega \subseteq J$. Indeed, for any inductive set K , ω can be defined as the intersection of all inductive subsets $J \subseteq K$. It can then be shown that this intersection ω lives in all inductive sets. This minimal inductive set ω is known as the **set of natural numbers**. It is easy to see that the famed axioms of Peano are satisfied by ω .

Intuitively, we'd want to say something like

$$\omega = \bigcap \{k \mid k \text{ is inductive}\}$$

The only problem is, a priori, the above intersection is not a set! This is why we need this axiom.

1.2 Transitive Sets

This is a quick recap section on transitive sets, ordinals, cardinals, and more.

Definition 1.2.1 (Transitive Set). A set A is **transitive** if $\forall y \in A [y \subseteq A \leftrightarrow \forall y \in A \forall x \in y (x \in A)]$

1.2.1 Ordinals

Definition 1.2.2 (Ordinal). A set α is an **ordinal** iff α is a transitive set and $(\alpha, \in)$ is a well-ordering.

Note that writing $(A, \in)$ really means the structure (A, R) where $R = \{(x, y) \in A \times A \mid x \in y\}$. It's slightly problematic to call $\in$ itself a relation because it isn't a set: it's (strictly speaking) just a symbol, which is interpreted as a set (a relation) in a model of set theory. We have to be careful with these sorts of things.

Example 1.2.3 (Examples of Ordinals).

1. $0 = \emptyset$, $1 = \{0\}$, $2 = \{0, 1\}$, $3 = \{0, 1, 2\}$, etc.
2. $\omega = \{0, 1, 2, \dots\}$
3. $\omega + 1 = \omega \cup \{\omega\}$, $\omega + 2$, etc.
4. $\omega + \omega$

and so on.

We will eventually prove this in more generality, but here's a useful theorem.

Theorem 1.2.4 (Mostowski Collapse Theorem for Well-Orderings). *For all well-orderings (A, R) , there is a unique ordinal α such that there is a unique isomorphism π from (A, R) to $(\alpha, \in)$ given by*

$$\pi(y) = \{\pi(x) \mid xRy\}$$

This is a special case of what's called (surprise surprise) the Mostowski Collapse Theorem. Some things we can do include removing the well/linear requirement on the ordering and replace “ordinal” with just “transitive set”.

What the theorem above says is that the ordinals form a complete set of representatives for the class of all well-orderings up to isomorphism.

1.3 The V-Hierarchy

Definition 1.3.1 (The V-Hierarchy). Define $V_0 = \emptyset$, $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ for any ordinal α , and $V_\beta = \bigcup_{\alpha < \beta} V_\alpha$ for any limit ordinal β .

One can show that $\alpha \leq \beta \rightarrow V_\alpha \subseteq V_\beta$.

We can ask ourselves what exactly ZFC is describing. The answer is, **the V-hierarchy**.

Interestingly,

$$V_\omega \models \mathbf{ZFC} - \text{infinity} + \neg \text{infinity}$$

Even more interestingly, ZFC doesn't prove its own consistency (if it could, it would be inconsistent, by Gödel), but it comes exactly one axiom away: it can prove the existence of models of everything except for the axiom of infinity.

1.3.1 Old stuff

It is worth asking ourselves the following question: *Why does V_ω exist?* Or, more precisely, *why is V_ω a set?*

By induction, we know that each V_n is a set. We want to essentially see that the map $n \mapsto V_n$ is a *set* (ie, a *function*). Then, if we take the union of the range of this function, whose domain is the set ω whose existence is given by ZFC Axiom 9, we can define V_ω to be that union.

We invoke ZFC Axiom 6, the Axiom of Replacement. Let $\varphi(x, y)$ be the formula capturing the following facts:

1. y is a function with domain $\text{dom}(y) = x \cup \{x\}$.
2. If $\emptyset \in \text{dom}(y)$, then $y(\emptyset) = \emptyset$.
3. If $z \cup \{z\} \in \text{dom}(y)$, then $z \in \text{dom}(y)$ and $y(z \cup \{z\}) = \mathcal{P}(z)$.

The idea is that $y = \langle V_0, V_1, V_2, \dots, V_{n+1} \rangle$ ¹ for some $n \in \omega$. Then,

$$\mathbf{ZFC} \vdash \forall n \in \omega \exists! y \varphi(n, y)$$

Applying the Axiom of Replacement (ZFC Axiom 6), we know that the map that takes any $n \in \omega$ to its unique witness y for n is, *indeed, a function*. In other words,

$$\langle \langle V_m \mid m \leq n \rangle \mid n < \omega \rangle$$

is a set. From this, we can read off

$$\langle V_n \mid n < \omega \rangle$$

by taking a union.

Remark. Recall that $(V_\omega, \in)$ is shorthand for (V_ω, R) , where $R = \{(x, y) \in V_\omega \times V_\omega \mid x \in y\}$. It is possible to show not only that

$$(V_\omega, \in) \models \text{ZFC without the Axiom of Infinity}$$

¹By this, we mean it is a *sequence*, ie, a function from ω to something that takes 0 to V_0 , 1 to V_1 , and so on.

but the stronger fact that

$$(V_\omega, \in) \models \text{ZFC without the Axiom of Infinity but with its } \textit{negation}$$

Thus,

$$\mathbf{ZFC} \vdash \text{ZFC without Infinity but with } \neg \text{Infinity is consistent}$$

ie, from ZFC, we can deduce that the entirety of ZFC, except with the negation of the axiom of infinity instead of the axiom itself, is consistent.

We note that the typical background setting for this course is the entirety of ZFC. However, as and when we need to either get rid of an axiom, or introduce a new one, we will mention it explicitly. We also use the following notational conventions.

Notation. We use **ZFC** to mean we have all the ZFC axioms. As and when we need variations, we do the following.

ZF := “ZFC without the Axiom of Choice”

ZFC – F := “ZFC without the Axiom of Foundation”

ZF – F := “ZF without the Axiom of Foundation”

Here, we end our discussion on the axioms of set theory, and proceed to consider a broader type of object than sets, known as classes.

1.4 Classes

Recall Russell’s Paradox: there is no set of all sets. One reason for this (which relies on the Axiom of Foundation) is that if V is the set of all sets, then $V \in V$, so V has no $\in$ -minimal member, which contradicts the Axiom of Foundation. Without the Axiom of Foundation, if V is the set of all sets, then if we set

$$W = \{x \in V \mid x \notin x\}$$

(which is a set by the Axiom of Comprehension), then if $W \in W$, then $W \notin W$, and if $W \notin W$, then $W \in W$, which is a contradiction, meaning there can be no set of all sets.

Nevertheless, we really do want to write things like

$$V = \{x \mid x \text{ is a set}\} = \{x \mid x = x\}$$

We would also like to write things like

$$\mathbf{OR} = \{\alpha \mid \alpha \text{ is an ordinal}\} \{V_\alpha \mid \alpha \in \mathbf{OR}\}$$

The only problem is, none of these are sets. Indeed, the only guarantee of the Axiom of Comprehension is that

$$\{x \in A \mid \varphi(x, \bar{y})\}$$

is a set; there is no guarantee whatsoever that

$$\{x \mid \varphi(x, \bar{y})\}$$

is a set. It sometimes is, but it isn't always.

There are two ways you can prove that $V = \{x \mid x = x\}$ is not a set. The first is that if it were, then $V \in V$ would hold, meaning it would have no $\in$ -minimal element, contradicting foundation. The second is that if it were, $W = \{x \in V \mid x \notin x\}$ would be a valid set, by comprehension. Then, $w \in W \leftrightarrow w \notin W$, a contradiction.

There are two perspectives we can take. The first is the **external perspective**: this is the perspective from which you would look at models of ZFC. There is also the internal perspective, where you are already inside a model of ZFC. Now this internal perspective also has a notion of logic and set theory inside of it. So there's formalisations of set theory inside models of set theory, and these formalisations have their own notions of models of set theory, and so it goes.

At some point, we will study these things in detail.

For the time being, we will use the term **class** to describe something we might want to express as a set, even if it cannot actually be a set for cardinality reasons.

Note that we would want sets to be classes. Indeed, anything of the form $\{x \mid \varphi(x)\}$, with $\varphi(x)$ a formula in the language of set theory, should be a class. Unfortunately, this is not generous enough: we want, for each set A , that

$$A = \{x \mid x \in A\} = \{x \mid \varphi(x, A)\}$$

to be included as well. In other words, we want to have formulae φ that not only allow us to substitute free variables with input sets x but also allow us to substitute more free variables with parameter sets A .

Formulae and parameters determine the class, but the converse is not true. In other words, if

$$\forall x (\varphi(x) \leftrightarrow \varphi'(x))$$

then

$$\{x \mid \varphi(x)\} = \{x \mid \varphi'(x)\}$$

For example, we know that

$$\varphi(x) \leftrightarrow (\varphi(x) \wedge x = x)$$

So you can define the same class using more than one formula.

1.4.1 The Class of Ordinals

Definition 1.4.1 (The Class of Ordinals). Define

$$\mathbf{OR} := \{\alpha \mid \alpha \text{ is an ordinal}\}$$

We can show that **OR** is a class.

We can then say that the ‘mapping’

$$\alpha \mapsto V_\alpha$$

is a **class function**: it takes elements of a class and maps them to elements of another class.

We can represent this mapping as

$$\{(\alpha, V_\alpha) \mid \alpha \in \mathbf{OR}\}$$

Now we can ask ourselves the following question: for which formulae φ is this set equal to $\{x \mid \varphi(x)\}$?

Let's give two such formulae based on the following.

Theorem 1.4.2. $\forall \gamma \in \mathbf{OR}, \forall S \in V$, *TFAE*:

1. $S = V_\gamma$
2. $\exists g$ a function with the properties that
 - (a) The domain of g is $\gamma + 1$
 - (b) $g(0) = \emptyset$
 - (c) $\forall \alpha < \gamma, g(\alpha + 1) = \mathcal{P}(g(\alpha))$
 - (d) $\forall \beta \leq \gamma$ with β a limit ordinal, $g(\beta) = \bigcup_{\alpha < \beta} g(\alpha)$
 - (e) $g(\gamma) = S$
3. Every function g exhibiting properties (a)-(d) above also exhibits property (e).

1.4.2 Well-Foundedness

We now define the class of well-founded sets.

Definition 1.4.3 (The Class of Well-Founded Sets). Define

$$\mathbf{WF} := \bigcup_{\alpha \in \mathbf{OR}} V_\alpha$$

That is, **WF** consists of all x such that there is some ordinal α with $x \in V_\alpha$.

WF is a class.

We will eventually see that the axiom of foundation is equivalent to saying that $V = \mathbf{WF}$ (under **ZF** without foundation).

Proposition 1.4.4. For all $\gamma \in \mathbf{OR}$,

- (1) $_{\gamma}$ $\forall \alpha \leq \gamma, V_{\alpha} \subseteq V_{\gamma}$
- (2) $_{\gamma}$ V_{γ} is a transitive set

Proof. We argue by induction on ordinals γ .

For the base step ($\gamma = 0$), both (1) $_0$ and (2) $_0$ are clear.

For the successor step, if $\gamma = \beta + 1$,

(1) $_{\gamma}$ Let $\alpha \leq \beta$ and $x \in V_{\alpha}$. By (1) $_{\beta}$, $x \in V_{\beta}$. By (2) $_{\beta}$, $x \subseteq V_{\beta}$. So, $x \in \mathcal{P}(V_{\beta}) = V_{\beta+1} = V_{\gamma}$.

(2) $_{\gamma}$ Let $y \in V_{\gamma} = \mathcal{P}(V_{\beta})$. Then, $y \subseteq V_{\beta}$. By (1) $_{\beta}$, $y \subseteq V_{\gamma}$.

Finally, for the limit step, ie, if γ is a limit ordinal, then (1) $_{\gamma}$ and (2) $_{\gamma}$ both follow from the fact that $\forall \beta < \gamma$, (1) $_{\beta}$ and (2) $_{\beta}$ both hold. (This is because in the limit case, constructing V_{γ} is just done by taking unions.) □

Definition 1.4.5 (Rank). Define, for $x \in \mathbf{WF}$, the **rank** of x , denoted $\text{rank}(x)$, to be the least $\alpha \in \mathbf{OR}$ such that $x \in V_{\alpha+1}$.

Indeed, this is equivalent to $\text{rank}(x)$ being the least $\alpha \in \mathbf{OR}$ such that $x \subseteq V_{\alpha}$.

Proposition 1.4.6. The mapping

$$x \mapsto \text{rank}(x)$$

is a class surjection from $\mathbf{WF}$ to $\mathbf{OR}$.

We won't prove this, but we'll say it's easy enough to prove and leave it at that.

Proposition 1.4.7. $\forall \beta \in \mathbf{OR}$,

$$V_{\beta} = \{x \in \mathbf{WF} \mid \text{rank}(x) < \beta\}$$

Proof sketch. You can just do induction on β and apply the definition of the rank (Definition 1.4.5). □

We can detect membership in **WF** one level down.

Proposition 1.4.8. $\forall y (y \in \mathbf{WF} \leftrightarrow y \subset \mathbf{WF})$

Proof.

$\rightarrow$ Let $\beta = \text{rank}(y)$. Then $y \subseteq V_\beta \subset \mathbf{WF}$.

$\leftarrow$ Assume $y \subset \mathbf{WF}$. Define $f : y \rightarrow \mathbf{OR}$ by $f(x) = \text{rank}(x) + 1$. We're using replacement to know that $f \in V$.

Let $\beta = \sup_{x \in y} f(x)$. Then, $y \subseteq V_\beta$. So $y \in \mathcal{P}(V_\beta) = V_{\beta+1} \subset \mathbf{WF}$. □

Passing to a member drops the rank strictly.

Proposition 1.4.9.

$$\forall y \in \mathbf{WF}, \forall x \in y, [x \in \mathbf{WF} \wedge \text{rank}(x) < \text{rank}(y)]$$

Proof. We know $x \in y \subseteq V_{\text{rank}(y)} \subset \mathbf{WF}$, so $x \in \mathbf{WF}$. This also shows that $\text{rank}(x) + 1 \leq \text{rank}(y)$, so $\text{rank}(x) < \text{rank}(y)$. □

This is enough to compute the rank of a set from the ranks of its members.

Proposition 1.4.10 (The Rank Formula).

$$\forall y \in \mathbf{WF}, \text{rank}(y) = \sup_{x \in y} (\text{rank}(x) + 1)$$

Proof. Put $\rho = \text{rank}(y)$ and $\sigma = \sup_{x \in y} (\text{rank}(x) + 1)$. By Proposition 1.4.9, if $x \in y$, then $\text{rank}(x) < \text{rank}(y) = \rho$, so $\text{rank}(x) + 1 \leq \rho$. So $\sigma \leq \rho$.

It suffices to show that $y \subseteq V_\sigma$ to conclude that $\rho \leq \sigma$. Now $V_\sigma = \{x \mid \text{rank}(x) < \sigma\}$. Every $x \in y$ has $\text{rank}(x) + 1 \leq \sigma$, by the definition of σ as a supremum, and so $\text{rank}(x) < \sigma$. So y is indeed a subset. $\square$

The ordinals sit inside **WF** in the simplest way imaginable.

Proposition 1.4.11.

$$\forall \beta \in \mathbf{OR}, [\beta \in \mathbf{WF} \wedge \text{rank}(\beta) = \beta]$$

Proof. We argue by induction on β . Base case is trivial. In the inductive case, $\beta \subseteq \mathbf{WF}$, so by previous prop, $\beta \in \mathbf{WF}$. So,

$$\begin{aligned} \text{rank}(\beta) &= \sup_{\alpha < \beta} (\text{rank}(\alpha) + 1) \\ &= \sup_{\alpha < \beta} (\alpha + 1) \\ &= \beta \end{aligned}$$

$\square$

It follows that each level of the hierarchy meets **OR** in exactly the ordinals below its index.

Proposition 1.4.12.

$$\forall \beta \in \mathbf{OR}, [V_\beta \cap \mathbf{OR} = \beta]$$

Proof. We know

$$\begin{aligned} V_\beta \cap \mathbf{OR} &= \{x \in \mathbf{WF} \mid \text{rank}(x) < \beta\} \cap \mathbf{OR} \\ &= \{\alpha \in \mathbf{OR} \mid \text{rank}(\alpha) < \beta\} \\ &= \{\alpha \in \mathbf{OR} \mid \alpha < \beta\} \\ &= \beta \end{aligned}$$

$\square$

1.5 Cardinals and the Axiom of Choice

We now ask how big the levels of the V -hierarchy are. At the bottom the answer is elementary; one level above V_ω it already depends on the Axiom of Choice, and that dependence is what the rest of this section is about.

1.5.1 The Size of V_ω

The finite levels of the hierarchy are as small as they could be.

Proposition 1.5.1. *For all $n < \omega$, the set V_n is finite.*

Indeed, their sizes are $0, 2^0, 2^{2^0}$, and so on.

So V_ω is the first level that is infinite, and it turns out to be no bigger than it has to be.

Proposition 1.5.2. *V_ω is countable.*

Proof using AC. Write

$$V_\omega = \bigcup_{n < \omega} V_n$$

AC tells us that a countable union of countable sets is countable. □

Proof that avoids AC. By recursion on $n < \omega$, we can define $R_n \subseteq V_n \times V_n$ so that

1. (V_n, R_n) is a well-ordering
2. R_{n+1} end-extends R_n

Then, we'll be able to put $R_\omega = \bigcup_{n < \omega} R_n$ and show that R_ω is a well-ordering of type ω .

For $x, y \in V_{n+1}$, put $x R_{n+1} y$ iff

1. $x, y \in V_n$ and $x R_n y$.
2. $x \in V_n$ and $y \in V_{n+1} \setminus V_n$.

3. $x, y \in V_{n+1} \setminus V_n$ and x precedes y in the ordering we are about to describe.

Since (V_n, R_n) is a well-ordering of a finite set, we may enumerate V_n as $a_0 R_n a_1 R_n \cdots R_n a_{k-1}$, where $k = |V_n|$. Every member of V_{n+1} is a subset of V_n , hence determined by which of the a_i it contains, so we may order V_{n+1} by binary numerals: given distinct $x, y \in V_{n+1}$, take the largest i for which a_i lies in the symmetric difference of x and y , and let x precede y iff $a_i \in y$. Case (3) is this ordering, restricted to $V_{n+1} \setminus V_n$.

So R_{n+1} is R_n followed by a linear ordering of the finitely many sets in $V_{n+1} \setminus V_n$, which makes it a well-ordering and gives us (1). It agrees with R_n on V_n and places all of V_n before all of $V_{n+1} \setminus V_n$, so V_n is an initial segment of (V_{n+1}, R_{n+1}) , which gives us (2).

The point of all this is that R_{n+1} is defined outright from R_n , with no choices made along the way, and the recursion starts from $R_0 = \emptyset$ on $V_0 = \emptyset$. The R_n increase, so R_ω linearly orders V_ω . Chaining (2) makes V_n an initial segment of (V_m, R_m) for every $m \geq n$, so each $x \in V_\omega$ has all of its R_ω -predecessors inside the finite V_n it came from, and every proper initial segment of R_ω is therefore finite. Since $\omega \subseteq V_\omega$, the ordering is infinite, so sending each x to its number of predecessors is an isomorphism onto ω . $\square$

1.5.2 Cardinality Without Choice

The reason for giving that second proof is that without AC, countable unions are genuinely badly behaved.

Remark. If ZF is consistent, then so is ZF + “there’s a countable union of pairs that has no cardinality”.

The same failure bites one level up the hierarchy.

Remark. Without AC, we don’t know whether the set $V_{\omega+1} = \mathcal{P}(V_\omega)$ even has a *cardinality*. With AC,

$$|V_{\omega+1}| = |\mathcal{P}(V_\omega)| = 2^{\aleph_0}$$

where $\aleph_0 = \omega = |V_\omega|$.

1.5.3 Cardinal Successors

Not everything is lost, though. The successor of a cardinal survives without choice, provided we go about it the right way.

Given AC, we know that $2^\kappa > \kappa$ and we can define κ^+ to be the least cardinal strictly greater than κ . Without AC, we still know, given a cardinal κ , that

$$\{R \subseteq \kappa \times \kappa \mid (\kappa, R) \text{ is a well-ordering}\}$$

is a set. We can still have Mostowski Collapse for well-orderings. We can then conclude that

$$\{\text{types}(\kappa, R) \mid (\kappa, R) \text{ is a well-ordering}\}$$

is a set of ordinals. Then, we can set κ^+ to simply be the supremum of this set.

1.5.4 The Alephs and the Beths

The point of the last subsection is that exactly one of the following two definitions needs the Axiom of Choice.

Without AC, define

$$\begin{aligned}\aleph_0 &:= \omega \\ \aleph_{\alpha+1} &:= (\aleph_\alpha)^+ \\ \aleph_\beta &:= \sup_{\alpha < \beta} (\aleph_\alpha) \text{ if } \beta \text{ is a limit ordinal}\end{aligned}$$

With AC, define

$$\begin{aligned}\beth_0 &:= \omega \\ \beth_{\alpha+1} &:= 2^{\beth_\alpha} \\ \beth_\beta &:= \sup_{\alpha < \beta} (\beth_\alpha) \text{ if } \beta \text{ is a limit ordinal}\end{aligned}$$

It's clear **sorry**

Chapter 2

Another Chapter

You get the idea.

2.1 Introducing the Main Object of Study in this Chapter

Woah. Very cool.

2.2 Another Section

Yup, \lipsum time. Boy do I love L^AT_EX!

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Bibliography

These lecture notes are based heavily on the following references:

- [1] Nicolas Bourbaki. *Groupes et Algèbres de Lie. Chapitre 1*. 1st ed. Éléments de Mathématique. Édition originale publiée par Hermann, Paris, 1972. Springer Berlin, Heidelberg, 2007. ISBN: 978-3-540-35337-9. DOI: <https://doi.org/10.1007/978-3-540-35337-9>.

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